

CLI Agent Task Contract Schema v0.1

Machine-readable task envelope for C-Governed CLI Agent Mesh operations

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Primary subject:	persistent c entities using CLI/cloud agents as bounded executable workers
Primary boundary:	no CLI agent may execute material work without declared scope, permissions, data policy, execution budget, output requirements, approval rules, and failure behavior.

Contents

0. Executive definition	5
1. Purpose	5
2. Non-goals	6
3. Corpus bridge set	7
3.1 Explicit bridge: $c = a + b$	7
3.2 Quiet bridge I: Ashby and operational variety	7
3.3 Quiet bridge II: information theory and leakage control	7
3.4 Earth paragraph	7
4. Core invariants	8
4.1 Required invariants	8
4.2 Minimal valid contract	8
5. Task risk classes	9
5.1 Risk escalation rule	9
6. Agent roles	9
6.1 Role restrictions	10
7. Contract object overview	10
8. Field definitions	11
8.1 schema_version	11
8.2 contract_id	11
8.3 task_id	11
8.4 title	11
8.5 created_at	11
8.6 expires_at	11
8.7 status	12
8.8 requested_by	12

8.9 governing_entity	12
8.10 human_anchor	12
8.11 agent	13
8.12 objective	13
8.13 assertion_class	13
8.14 scope	13
8.15 data_policy	14
8.16 network_policy	14
8.17 execution	15
8.18 output_required	15
8.19 approval	16
8.20 witness	16
8.21 memory_gate	16
8.22 failure_behavior	17
8.23 red_lines	17
8.24 review	17
8.25 integrity	18
8.26 notes	18
9. Normative JSON Schema	18
10. Semantic validation rules	29
10.1 Red-line rule	29
10.2 External target rule	29
10.3 Role/action consistency rule	29
10.4 Risk/control consistency rule	29
10.5 Cloud data rule	29
10.6 Self-approval rule	30
10.7 Direct memory write rule	30
10.8 Network rule	30

10.9 Evidence preservation rule	30
10.10 Expiry rule	30
11. Example contracts	30
11.1 Read-only corpus review	30
11.2 Executor patch in sandbox	34
11.3 Invalid offensive contract	37
12. Contract validation workflow	40
13. Default failure mapping	41
14. Implementation notes	41
14.1 Canonicalization	41
14.2 Contract immutability	41
14.3 Append-only review	41
14.4 Human-readable pair	42
15. Conformance levels	42
16. Open issues	42
17. Closing rule	43

0. Executive definition

CLI Agent Task Contract Schema defines the machine-readable envelope that governs one bounded task assigned to a CLI/cloud agent under *c* governance.

A task contract answers:

```
Who requested this task?
Which c owns the task?
Which agent is assigned?
What may the agent read?
What may the agent write?
What may the agent execute?
What data is prohibited?
May the agent use the network?
What budget applies?
What output is required?
Who reviews?
What is witnessed?
What happens on failure?
```

The schema exists to prevent executable workers from operating through vague prompts, hidden permissions, silent defaults, or post-hoc justification.

Compact formula:

```
No task without contract.
No contract without scope.
No scope without denial rules.
No privileged transition without witness.
```

1. Purpose

CLI agents are not ordinary text responders. They may inspect files, edit repositories, run tests, create artifacts, change configuration, call tools, and influence operational state.

Therefore, a CLI agent task must be governed before execution.

This schema provides:

1. a canonical task envelope;
2. required identity and provenance fields;
3. role and capability binding;
4. allowed and denied paths;

5. command boundaries;
 6. data and privacy policy;
 7. network restrictions;
 8. sandbox/worktree requirements;
 9. L4 execution budgets;
 10. output requirements;
 11. approval rules;
 12. witness hooks;
 13. memory gate handling;
 14. fail-closed behavior;
 15. conformance validation rules.
-

2. Non-goals

This schema does not define or permit:

1. autonomous retaliation;
2. offensive cyber operations;
3. unauthorized access;
4. exploitation of live third-party systems;
5. credential theft;
6. malware behavior;
7. covert persistence;
8. evasion;
9. uncontrolled network scanning;
10. unbounded background operation;
11. direct write access to c core memory;
12. direct modification of identity, privilege, witness, or continuity core;
13. self-approval by the same agent that executed the work.

A valid task contract cannot authorize a prohibited action.

If a task attempts to encode a prohibited action, the contract is invalid and **MUST** resolve to `deny_and_quarantine`.

3. Corpus bridge set

3.1 Explicit bridge: $c = a + b$

In $c = a + b$, CLI agents belong to b : procedures, models, tools, interfaces, compute, and infrastructure.

They are not a .

They are not c .

They are bounded executable workers inside the technological substrate.

The task contract is the local law that binds one such worker to one task under c governance.

3.2 Quiet bridge I: Ashby and operational variety

A c operating in a complex technical environment needs multiple worker roles: reader, executor, tester, auditor, sentinel, archivist, judge-assistant. Requisite variety is useful only when each worker's channel is bounded. The task contract increases operational variety without surrendering control.

3.3 Quiet bridge II: information theory and leakage control

A CLI task is an information channel. It can transform local files, private context, logs, secrets, prompts, diffs, tests, and reports into new artifacts. The task contract reduces uncontrolled leakage by declaring what data may enter, what output may leave, and what material is prohibited.

3.4 Earth paragraph

A work order on a real site does not say "fix the building somehow." It names the room, the circuit, the materials, the responsible person, the lockout condition, the inspection requirement, and the sign-off path. If an electrician opens a panel outside the work order, the issue is not creativity; it is a safety violation. CLI agents need the same discipline. A vague prompt is not a permit to touch the whole system.

4. Core invariants

4.1 Required invariants

ID	Invariant	Requirement
TC-I01	Contract before execution	Material agent work MUST have a task contract.
TC-I02	Explicit owner	The governing c or human anchor MUST be identified.
TC-I03	Explicit agent role	The agent role MUST be declared before execution.
TC-I04	Deny-by-default	Missing permission means denied.
TC-I05	Allowed paths are not enough	Denied paths MUST also be declared for material tasks.
TC-I06	Secrets denied by default	Secret access MUST be false unless narrowly authorized.
TC-I07	Cloud upload denied by default	Cloud upload MUST be false unless explicitly authorized.
TC-I08	Sandbox first	Write or execute tasks MUST require sandbox/worktree/branch isolation.
TC-I09	Network bounded	Network mode MUST be none or allowlist; unrestricted network is invalid.
TC-I10	No self-approval	self_approval_allowed MUST be false.
TC-I11	Witness for privilege	Privileged transitions MUST require witness.
TC-I12	Memory gate	Agent output MUST NOT write directly into c memory.
TC-I13	Failure is specified	Failure behavior MUST be declared.
TC-I14	Rollback is specified	Write tasks MUST include rollback requirements.
TC-I15	Red lines override contract	Prohibited actions remain prohibited even if the contract says otherwise.

4.2 Minimal valid contract

A minimally valid material task contract **MUST** include:

```

schema_version
task_id
requested_by
governing_entity
agent.role
agent.agent_id
objective
risk_class
scope.allowed_paths
scope.denied_paths
data_policy
network_policy
execution
output_required
approval
failure_behavior

```


5. Task risk classes

Risk class	Meaning	Required control
R0	Read-only, harmless, public material	task contract recommended
R1	Low-risk documentation or formatting	contract + allowed paths
R2	Code, schema, build, or data transformation	sandbox + tests + reviewer
R3	Release, publication, metadata, public surface	witness + c gate
R4	Memory, identity, privilege, witness, continuity, or agent governance impact	witness + c gate + human gate
R5	Incident, legal, security-sensitive, or evidence-sensitive action	preserve evidence + human gate + restricted data policy
RX	Prohibited, offensive, unauthorized, or uncontrolled action	deny + quarantine

5.1 Risk escalation rule

If multiple risk classes apply, the highest class controls.

Example:

A documentation change touching public release metadata is R3, not R1.

A code fix touching privilege checks is R4, not R2.

An incident task involving logs and tokens is R5, not R2.

6. Agent roles

The contract **MUST** bind the task to one primary agent role.

Allowed roles:

```
reader
executor
tester
auditor
archivist
sentinel
judge_assistant
orchestrator_limited
```

6.1 Role restrictions

Role	May write?	May execute?	May approve?	Notes
reader	no	no	no	read / summarize / compare only
executor	sandbox/branch only	bounded	no	produces patch/diff
tester	no or test-output only	tests only	no	may run allowed validation
auditor	no	no or read-only checks	no	reviews another agent's output
archivist	controlled metadata only	validation only	no	indexes, hashes, release-control prep
sentinel	no by default	monitoring only	no	detects drift/anomaly
judge_assistant	no	no	no	advises c, does not decide
orchestrator_limited	no direct task mutation	dispatch only	no	routes tasks, cannot grant new privilege silently

7. Contract object overview

Canonical object name:

```
CLI_AGENT_TASK_CONTRACT
```

Canonical top-level fields:

```
schema_version: string
contract_id: string
task_id: string
title: string
created_at: string
expires_at: string | null
status: draft | active | completed | failed | held | frozen | quarantined | revoked
requested_by: c | human_anchor | scheduled_policy | incident_policy
governing_entity: object
human_anchor: object | null
agent: object
objective: string
risk_class: R0 | R1 | R2 | R3 | R4 | R5 | RX
assertion_class: string
scope: object
data_policy: object
network_policy: object
execution: object
output_required: array
approval: object
witness: object
```

```
memory_gate: object
failure_behavior: object
red_lines: object
review: object
integrity: object
notes: object
```

8. Field definitions

8.1 `schema_version`

Required string.

Must equal:

```
cli-agent-task-contract-0.1
```

8.2 `contract_id`

Required string.

Stable identifier for this task contract.

Recommended format:

```
catc-YYYYMMDD-HHMMSS-shortslug
```

8.3 `task_id`

Required string.

Identifier used by the agent execution system.

8.4 `title`

Required string.

Human-readable task title.

8.5 `created_at`

Required timestamp.

ISO-8601 UTC recommended.

8.6 `expires_at`

Optional timestamp or null.

If expired, the agent **MUST** stop unless a new contract is issued.

8.7 status

Required enum:

```
draft
active
completed
failed
held
frozen
quarantined
revoked
```

8.8 requested_by

Required enum:

```
c
human_anchor
scheduled_policy
incident_policy
```

8.9 governing_entity

Required object.

Fields:

```
governing_entity:
  entity_id: string
  entity_name: string
  continuity_ref: string | null
  authority_scope: local | repository | project | incident | release | memory | other
```

8.10 human_anchor

Optional object or null.

Required for R4, R5, and all high-risk approvals.

```
human_anchor:
  anchor_id: string
  display_name: string
  approval_required: boolean
  legal_review_required: boolean
```

8.11 agent

Required object.

```
agent:
  agent_id: string
  agent_name: string
  provider: local | openai | google | anthropic | other | unknown
  role: reader | executor | tester | auditor | archivist | sentinel | judge_assistant |
    orchestrator_limited
  runtime: local_cli | cloud_cli | api_agent | container_agent | hybrid
  version: string | null
  capability_profile_ref: string | null
  trust_level: untrusted | provisional | trusted_limited | trusted_high
```

A high trust level does not remove scope requirements.

8.12 objective

Required string.

Must describe the task outcome in bounded terms.

Bad:

```
Fix everything.
```

Good:

```
Validate Markdown links under docs/cli-agent/ and produce a patch only for broken internal
links.
```

8.13 assertion_class

Required string.

Recommended values:

```
C-A4
C-A7
C-A10
```

Other values may be used if defined by the parent corpus.

8.14 scope

Required object.

```
scope:
```

```
repository: string | null
branch_or_worktree: string | null
allowed_paths:
  - string
denied_paths:
  - string
allowed_commands:
  - string
denied_commands:
  - string
allowed_file_globs:
  - string
denied_file_globs:
  - string
max_files_touched: integer | null
max_diff_lines: integer | null
external_targets_allowed: false
external_targets:
  - string
```

`external_targets_allowed` MUST be false unless the task is explicitly authorized and lawful. For this protocol v0.1, live third-party offensive testing remains prohibited.

8.15 data_policy

Required object.

```
data_policy:
  classification: public | internal | private | restricted | sealed | legal_sensitive |
    incident_sensitive
  secrets_allowed: false
  private_memory_allowed: false
  sealed_material_allowed: false
  legal_privileged_material_allowed: false
  child_data_allowed: false
  raw_witness_evidence_allowed: false
  cloud_upload_allowed: false
  prompt_minimization_required: true
  redaction_required: true
  retention: ephemeral | operational | audit | legal_hold
  output_sanitization_required: true
```

8.16 network_policy

Required object.

```
network_policy:
```

```
mode: none | allowlist
allowed_endpoints:
  - string
denied_endpoints:
  - string
package_install_allowed: false
external_fetch_allowed: false
```

Unrestricted network is not a valid mode in v0.1.

8.17 execution

Required object.

```
execution:
  sandbox_required: true
  sandbox_id: string | null
  branch_required: boolean
  container_required: boolean
  dry_run_first: boolean
  max_runtime_minutes: integer
  max_retries: integer
  max_cost_eur: number | null
  max_tokens: integer | null
  stop_on_scope_violation: true
  preserve_evidence_before_repair: boolean
  rollback_required_if_partial_write: boolean
```

8.18 output_required

Required array of enums.

Allowed values:

```
summary
changed_files
diff
tests_run
test_results
risk_report
rollback_plan
unresolved_questions
scope_report
commands_run
hashes
witness_refs
memory_gate_recommendation
```

8.19 approval

Required object.

```
approval:
  self_approval_allowed: false
  reviewer_required: true
  reviewer_agent_id: string | null
  c_gate_required: true
  human_gate_required: boolean
  legal_review_required: boolean
  merge_allowed_without_human: false
  publish_allowed_without_human: false
  deploy_allowed_without_human: false
```

For R4 and R5, human_gate_required MUST be true.

8.20 witness

Required object.

```
witness:
  witness_required: boolean
  witness_profile: none | operational | privileged | incident | release | memory_gate
  witness_event_families:
    - cli_agent.connection
    - cli_agent.permission
    - cli_agent.execution
    - cli_agent.review
    - cli_agent.memory_gate
    - cli_agent.incident
    - cli_agent.anomaly
  append_only_required: true
  hash_required: boolean
  signature_required: boolean
```

Privileged transitions require witness.

8.21 memory_gate

Required object.

```
memory_gate:
  direct_memory_write_allowed: false
  memory_proposal_allowed: boolean
  proposed_memory_class: MG-0 | MG-1 | MG-2 | MG-3 | MG-4 | MG-Q | null
```



```
c_review_required: true
uncertainty_required: true
quarantine_on_uncertainty: boolean
```

8.22 failure_behavior

Required object.

```
failure_behavior:
  default_on_failure: hold | freeze | quarantine | rollback | deny_and_quarantine
  default_on_scope_violation: quarantine
  default_on_secret_exposure: freeze_and_escalate
  default_on_network_violation: quarantine
  default_on_missing_witness: hold
  default_on_agent_self_approval: revoke_and_quarantine
  rollback_strategy: none | revert_patch | restore_snapshot | manual_review
```

8.23 red_lines

Required object.

All fields MUST be false.

```
red_lines:
  offensive_action_requested: false
  live_external_exploitation: false
  hack_back: false
  credential_theft: false
  malware_behavior: false
  covert_persistence: false
  evasion: false
  destructive_action: false
  autonomous_retaliation: false
  third_party_scanning: false
```

If any red-line field is true, the task is invalid.

8.24 review

Required object.

```
review:
  independent_reviewer_required: boolean
  executor_reviewer_separation_required: boolean
  quorum_required: boolean
  quorum_profile: none | codex_gemini_local | executor_tester_auditor | custom
  disagreement_policy: accept_if_minor | hold | c_review | human_review | arl_review
```

8.25 integrity

Required object.

```
integrity:
  input_hash: string | null
  contract_hash: string | null
  expected_output_hash: string | null
  canonicalization: none | json_canonicalization | yaml_canonicalization
  previous_contract_ref: string | null
  supersedes_contract_ref: string | null
```

8.26 notes

Optional object.

```
notes:
  human_notes: string | null
  c_notes: string | null
  unresolved_issues:
    - string
  assumptions:
    - string
```

9. Normative JSON Schema

The following JSON Schema is the canonical v0.1 structural schema.

```
{
  "$schema": "https://json-schema.org/draft/2020-12/schema",
  "$id": "https://example.local/schemas/cli-agent-task-contract-0.1.schema.json",
  "title": "CLI Agent Task Contract Schema v0.1",
  "type": "object",
  "additionalProperties": false,
  "required": [
    "schema_version",
    "contract_id",
    "task_id",
    "title",
    "created_at",
    "status",
    "requested_by",
    "governing_entity",
    "agent",
    "objective",
  ]
}
```

```
    "risk_class",
    "assertion_class",
    "scope",
    "data_policy",
    "network_policy",
    "execution",
    "output_required",
    "approval",
    "witness",
    "memory_gate",
    "failure_behavior",
    "red_lines",
    "review",
    "integrity"
  ],
  "properties": {
    "schema_version": {
      "const": "cli-agent-task-contract-0.1"
    },
    "contract_id": {
      "type": "string",
      "minLength": 8
    },
    "task_id": {
      "type": "string",
      "minLength": 3
    },
    "title": {
      "type": "string",
      "minLength": 3
    },
    "created_at": {
      "type": "string",
      "format": "date-time"
    },
    "expires_at": {
      "type": ["string", "null"],
      "format": "date-time"
    },
    "status": {
      "enum": ["draft", "active", "completed", "failed", "held", "frozen", "quarantined",
        "revoked"]
    },
    "requested_by": {
      "enum": ["c", "human_anchor", "scheduled_policy", "incident_policy"]
    }
  }
}
```

```
},
"governing_entity": {
  "type": "object",
  "additionalProperties": false,
  "required": ["entity_id", "entity_name", "authority_scope"],
  "properties": {
    "entity_id": { "type": "string", "minLength": 1 },
    "entity_name": { "type": "string", "minLength": 1 },
    "continuity_ref": { "type": ["string", "null"] },
    "authority_scope": {
      "enum": ["local", "repository", "project", "incident", "release", "memory", "other"]
    }
  }
},
"human_anchor": {
  "type": ["object", "null"],
  "additionalProperties": false,
  "required": ["anchor_id", "display_name", "approval_required", "legal_review_required"],
  "properties": {
    "anchor_id": { "type": "string" },
    "display_name": { "type": "string" },
    "approval_required": { "type": "boolean" },
    "legal_review_required": { "type": "boolean" }
  }
},
"agent": {
  "type": "object",
  "additionalProperties": false,
  "required": ["agent_id", "agent_name", "provider", "role", "runtime", "trust_level"],
  "properties": {
    "agent_id": { "type": "string", "minLength": 1 },
    "agent_name": { "type": "string", "minLength": 1 },
    "provider": {
      "enum": ["local", "openai", "google", "anthropic", "other", "unknown"]
    },
    "role": {
      "enum": ["reader", "executor", "tester", "auditor", "archivist", "sentinel",
        "judge_assistant", "orchestrator_limited"]
    },
    "runtime": {
      "enum": ["local_cli", "cloud_cli", "api_agent", "container_agent", "hybrid"]
    },
    "version": { "type": ["string", "null"] },
    "capability_profile_ref": { "type": ["string", "null"] },
    "trust_level": {
```

```
        "enum": ["untrusted", "provisional", "trusted_limited", "trusted_high"]
      }
    },
    "objective": {
      "type": "string",
      "minLength": 10
    },
    "risk_class": {
      "enum": ["R0", "R1", "R2", "R3", "R4", "R5", "RX"]
    },
    "assertion_class": {
      "type": "string",
      "minLength": 2
    },
    "scope": {
      "type": "object",
      "additionalProperties": false,
      "required": ["allowed_paths", "denied_paths", "allowed_commands", "denied_commands",
        "external_targets_allowed"],
      "properties": {
        "repository": { "type": ["string", "null"] },
        "branch_or_worktree": { "type": ["string", "null"] },
        "allowed_paths": {
          "type": "array",
          "items": { "type": "string" },
          "minItems": 1
        },
        "denied_paths": {
          "type": "array",
          "items": { "type": "string" },
          "minItems": 1
        },
        "allowed_commands": {
          "type": "array",
          "items": { "type": "string" }
        },
        "denied_commands": {
          "type": "array",
          "items": { "type": "string" }
        },
        "allowed_file_globs": {
          "type": "array",
          "items": { "type": "string" }
        }
      }
    },
```

```
    "denied_file_globs": {
      "type": "array",
      "items": { "type": "string" }
    },
    "max_files_touched": { "type": ["integer", "null"], "minimum": 0 },
    "max_diff_lines": { "type": ["integer", "null"], "minimum": 0 },
    "external_targets_allowed": { "const": false },
    "external_targets": {
      "type": "array",
      "items": { "type": "string" }
    }
  }
},
"data_policy": {
  "type": "object",
  "additionalProperties": false,
  "required": [
    "classification",
    "secrets_allowed",
    "private_memory_allowed",
    "sealed_material_allowed",
    "legal_privileged_material_allowed",
    "child_data_allowed",
    "raw_witness_evidence_allowed",
    "cloud_upload_allowed",
    "prompt_minimization_required",
    "redaction_required",
    "retention",
    "output_sanitization_required"
  ],
  "properties": {
    "classification": {
      "enum": ["public", "internal", "private", "restricted", "sealed", "legal_sensitive",
        "incident_sensitive"]
    },
    "secrets_allowed": { "type": "boolean" },
    "private_memory_allowed": { "type": "boolean" },
    "sealed_material_allowed": { "type": "boolean" },
    "legal_privileged_material_allowed": { "type": "boolean" },
    "child_data_allowed": { "type": "boolean" },
    "raw_witness_evidence_allowed": { "type": "boolean" },
    "cloud_upload_allowed": { "type": "boolean" },
    "prompt_minimization_required": { "type": "boolean" },
    "redaction_required": { "type": "boolean" },
    "retention": {
```

```
    "enum": ["ephemeral", "operational", "audit", "legal_hold"]
  },
  "output_sanitization_required": { "type": "boolean" }
}
},
"network_policy": {
  "type": "object",
  "additionalProperties": false,
  "required": ["mode", "allowed_endpoints", "denied_endpoints", "package_install_allowed",
    "external_fetch_allowed"],
  "properties": {
    "mode": { "enum": ["none", "allowlist"] },
    "allowed_endpoints": {
      "type": "array",
      "items": { "type": "string" }
    },
    "denied_endpoints": {
      "type": "array",
      "items": { "type": "string" }
    },
    "package_install_allowed": { "type": "boolean" },
    "external_fetch_allowed": { "type": "boolean" }
  }
},
"execution": {
  "type": "object",
  "additionalProperties": false,
  "required": [
    "sandbox_required",
    "branch_required",
    "container_required",
    "dry_run_first",
    "max_runtime_minutes",
    "max_retries",
    "stop_on_scope_violation",
    "preserve_evidence_before_repair",
    "rollback_required_if_partial_write"
  ],
  "properties": {
    "sandbox_required": { "type": "boolean" },
    "sandbox_id": { "type": ["string", "null"] },
    "branch_required": { "type": "boolean" },
    "container_required": { "type": "boolean" },
    "dry_run_first": { "type": "boolean" },
    "max_runtime_minutes": { "type": "integer", "minimum": 1 },
```

```
    "max_retries": { "type": "integer", "minimum": 0, "maximum": 10 },
    "max_cost_eur": { "type": ["number", "null"], "minimum": 0 },
    "max_tokens": { "type": ["integer", "null"], "minimum": 0 },
    "stop_on_scope_violation": { "type": "boolean" },
    "preserve_evidence_before_repair": { "type": "boolean" },
    "rollback_required_if_partial_write": { "type": "boolean" }
  }
},
"output_required": {
  "type": "array",
  "items": {
    "enum": [
      "summary",
      "changed_files",
      "diff",
      "tests_run",
      "test_results",
      "risk_report",
      "rollback_plan",
      "unresolved_questions",
      "scope_report",
      "commands_run",
      "hashes",
      "witness_refs",
      "memory_gate_recommendation"
    ]
  },
  "minItems": 1,
  "uniqueItems": true
},
"approval": {
  "type": "object",
  "additionalProperties": false,
  "required": [
    "self_approval_allowed",
    "reviewer_required",
    "c_gate_required",
    "human_gate_required",
    "legal_review_required",
    "merge_allowed_without_human",
    "publish_allowed_without_human",
    "deploy_allowed_without_human"
  ],
  "properties": {
    "self_approval_allowed": { "const": false },
```



```
    "reviewer_required": { "type": "boolean" },
    "reviewer_agent_id": { "type": ["string", "null"] },
    "c_gate_required": { "type": "boolean" },
    "human_gate_required": { "type": "boolean" },
    "legal_review_required": { "type": "boolean" },
    "merge_allowed_without_human": { "type": "boolean" },
    "publish_allowed_without_human": { "type": "boolean" },
    "deploy_allowed_without_human": { "type": "boolean" }
  }
},
"witness": {
  "type": "object",
  "additionalProperties": false,
  "required": ["witness_required", "witness_profile", "witness_event_families",
    "append_only_required", "hash_required", "signature_required"],
  "properties": {
    "witness_required": { "type": "boolean" },
    "witness_profile": {
      "enum": ["none", "operational", "privileged", "incident", "release", "memory_gate"]
    },
    "witness_event_families": {
      "type": "array",
      "items": {
        "enum": [
          "cli_agent.connection",
          "cli_agent.permission",
          "cli_agent.execution",
          "cli_agent.review",
          "cli_agent.memory_gate",
          "cli_agent.incident",
          "cli_agent.anomaly"
        ]
      }
    },
    "uniqueItems": true
  },
  "append_only_required": { "type": "boolean" },
  "hash_required": { "type": "boolean" },
  "signature_required": { "type": "boolean" }
}
},
"memory_gate": {
  "type": "object",
  "additionalProperties": false,
  "required": ["direct_memory_write_allowed", "memory_proposal_allowed",
    "c_review_required", "uncertainty_required", "quarantine_on_uncertainty"],
```

```
"properties": {
  "direct_memory_write_allowed": { "const": false },
  "memory_proposal_allowed": { "type": "boolean" },
  "proposed_memory_class": {
    "type": ["string", "null"],
    "enum": ["MG-0", "MG-1", "MG-2", "MG-3", "MG-4", "MG-Q", null]
  },
  "c_review_required": { "type": "boolean" },
  "uncertainty_required": { "type": "boolean" },
  "quarantine_on_uncertainty": { "type": "boolean" }
},
"failure_behavior": {
  "type": "object",
  "additionalProperties": false,
  "required": [
    "default_on_failure",
    "default_on_scope_violation",
    "default_on_secret_exposure",
    "default_on_network_violation",
    "default_on_missing_witness",
    "default_on_agent_self_approval",
    "rollback_strategy"
  ],
  "properties": {
    "default_on_failure": { "enum": ["hold", "freeze", "quarantine", "rollback",
    "deny_and_quarantine"] },
    "default_on_scope_violation": { "enum": ["hold", "freeze", "quarantine", "rollback",
    "deny_and_quarantine"] },
    "default_on_secret_exposure": { "enum": ["freeze_and_escalate", "quarantine",
    "deny_and_quarantine"] },
    "default_on_network_violation": { "enum": ["hold", "quarantine", "deny_and_quarantine"]
    },
    "default_on_missing_witness": { "enum": ["hold", "freeze", "quarantine"] },
    "default_on_agent_self_approval": { "enum": ["revoke_and_quarantine",
    "deny_and_quarantine"] },
    "rollback_strategy": { "enum": ["none", "revert_patch", "restore_snapshot",
    "manual_review"] }
  },
},
"red_lines": {
  "type": "object",
  "additionalProperties": false,
  "required": [
    "offensive_action_requested",
```

```
    "live_external_exploitation",
    "hack_back",
    "credential_theft",
    "malware_behavior",
    "covert_persistence",
    "evasion",
    "destructive_action",
    "autonomous_retaliation",
    "third_party_scanning"
  ],
  "properties": {
    "offensive_action_requested": { "const": false },
    "live_external_exploitation": { "const": false },
    "hack_back": { "const": false },
    "credential_theft": { "const": false },
    "malware_behavior": { "const": false },
    "covert_persistence": { "const": false },
    "evasion": { "const": false },
    "destructive_action": { "const": false },
    "autonomous_retaliation": { "const": false },
    "third_party_scanning": { "const": false }
  }
},
"review": {
  "type": "object",
  "additionalProperties": false,
  "required": ["independent_reviewer_required", "executor_reviewer_separation_required",
    "quorum_required", "quorum_profile", "disagreement_policy"],
  "properties": {
    "independent_reviewer_required": { "type": "boolean" },
    "executor_reviewer_separation_required": { "type": "boolean" },
    "quorum_required": { "type": "boolean" },
    "quorum_profile": { "enum": ["none", "codex_gemini_local", "executor_tester_auditor",
      "custom" ] },
    "disagreement_policy": { "enum": ["accept_if_minor", "hold", "c_review", "human_review",
      "arl_review" ] }
  }
},
"integrity": {
  "type": "object",
  "additionalProperties": false,
  "required": ["canonicalization"],
  "properties": {
    "input_hash": { "type": ["string", "null" ] },
    "contract_hash": { "type": ["string", "null" ] },
```

```
    "expected_output_hash": { "type": ["string", "null"] },
    "canonicalization": { "enum": ["none", "json_canonicalization", "yaml_canonicalization"]
    },
    "previous_contract_ref": { "type": ["string", "null"] },
    "supersedes_contract_ref": { "type": ["string", "null"] }
  }
},
"notes": {
  "type": "object",
  "additionalProperties": false,
  "properties": {
    "human_notes": { "type": ["string", "null"] },
    "c_notes": { "type": ["string", "null"] },
    "unresolved_issues": {
      "type": "array",
      "items": { "type": "string" }
    },
    "assumptions": {
      "type": "array",
      "items": { "type": "string" }
    }
  }
}
},
"allOf": [
  {
    "if": {
      "properties": { "risk_class": { "enum": ["R4", "R5"] } },
      "required": ["risk_class"]
    },
    "then": {
      "properties": {
        "approval": {
          "properties": {
            "human_gate_required": { "const": true }
          }
        }
      }
    }
  },
  {
    "if": {
      "properties": { "risk_class": { "const": "RX" } },
      "required": ["risk_class"]
    }
  },
]
```

```
    "then": {
      "properties": {
        "status": { "enum": ["held", "frozen", "quarantined", "revoked"] }
      }
    }
  }
}
```

10. Semantic validation rules

JSON Schema structural validation is necessary but not sufficient.

The following semantic rules **MUST** also be applied.

10.1 Red-line rule

If any red-line field is true, the contract is invalid regardless of all other fields.

10.2 External target rule

`external_targets_allowed` **MUST** remain false for ordinary protocol use.

A task involving external systems must be documented as owned, authorized, or delegated. Offensive or unauthorized use remains invalid.

10.3 Role/action consistency rule

A reader task **MUST NOT** include write commands.

An auditor task **MUST NOT** silently modify the work it audits.

A tester task **MUST NOT** alter tests to hide failures.

10.4 Risk/control consistency rule

- R2 requires sandbox or worktree isolation.
- R3 requires witness and c gate.
- R4 requires witness, c gate, and human gate.
- R5 requires evidence preservation before repair unless immediate containment requires otherwise.
- RX must resolve to denial/quarantine.

10.5 Cloud data rule

If `agent.runtime` is `cloud_cli`, then sensitive data fields **SHOULD** remain false unless explicitly justified:

```
secrets_allowed
private_memory_allowed
sealed_material_allowed
legal_privileged_material_allowed
child_data_allowed
raw_witness_evidence_allowed
cloud_upload_allowed
```

10.6 Self-approval rule

`self_approval_allowed` must be false.

No exception is valid in v0.1.

10.7 Direct memory write rule

`direct_memory_write_allowed` must be false.

No exception is valid in v0.1.

10.8 Network rule

`network_policy.mode` must be none or allowlist.

A task requiring broader network access must be decomposed into smaller scoped tasks.

10.9 Evidence preservation rule

For incident tasks, evidence preservation must happen before repair where possible.

10.10 Expiry rule

If `expires_at` is reached, the contract is no longer active and the agent must stop or request renewal.

11. Example contracts

11.1 Read-only corpus review

```
schema_version: cli-agent-task-contract-0.1
contract_id: catc-20260516-120000-corpus-review
task_id: task-corpus-review-001
title: Review CLI protocol documents for duplication
created_at: "2026-05-16T12:00:00Z"
expires_at: "2026-05-16T14:00:00Z"
status: active
requested_by: c
governing_entity:
```

```
entity_id: ester
entity_name: Ester
continuity_ref: null
authority_scope: project
human_anchor: null
agent:
  agent_id: gemini-reader-01
  agent_name: Gemini Reader
  provider: google
  role: reader
  runtime: cloud_cli
  version: null
  capability_profile_ref: null
  trust_level: provisional
objective: Review authorized CLI protocol drafts for duplication, contradiction, and missing
  boundaries without editing files.
risk_class: R1
assertion_class: C-A10
scope:
  repository: cli-agent-protocols
  branch_or_worktree: null
  allowed_paths:
    - docs/cli-agent/
  denied_paths:
    - secrets/
    - memory_core/
    - legal/
  allowed_commands: []
  denied_commands:
    - rm
    - git push
    - curl
  allowed_file_globs:
    - "*.md"
  denied_file_globs:
    - "*.env"
    - "*.key"
  max_files_touched: 0
  max_diff_lines: 0
  external_targets_allowed: false
  external_targets: []
data_policy:
  classification: internal
  secrets_allowed: false
  private_memory_allowed: false
```

```
sealed_material_allowed: false
legal_privileged_material_allowed: false
child_data_allowed: false
raw_witness_evidence_allowed: false
cloud_upload_allowed: false
prompt_minimization_required: true
redaction_required: true
retention: operational
output_sanitization_required: true
network_policy:
  mode: none
  allowed_endpoints: []
  denied_endpoints: []
  package_install_allowed: false
  external_fetch_allowed: false
execution:
  sandbox_required: false
  sandbox_id: null
  branch_required: false
  container_required: false
  dry_run_first: true
  max_runtime_minutes: 30
  max_retries: 1
  max_cost_eur: 1
  max_tokens: 100000
  stop_on_scope_violation: true
  preserve_evidence_before_repair: false
  rollback_required_if_partial_write: false
output_required:
  - summary
  - risk_report
  - unresolved_questions
  - scope_report
approval:
  self_approval_allowed: false
  reviewer_required: false
  reviewer_agent_id: null
  c_gate_required: true
  human_gate_required: false
  legal_review_required: false
  merge_allowed_without_human: false
  publish_allowed_without_human: false
  deploy_allowed_without_human: false
witness:
  witness_required: false
```



```
witness_profile: operational
witness_event_families:
  - cli_agent.execution
append_only_required: true
hash_required: false
signature_required: false
memory_gate:
  direct_memory_write_allowed: false
  memory_proposal_allowed: true
  proposed_memory_class: MG-2
  c_review_required: true
  uncertainty_required: true
  quarantine_on_uncertainty: true
failure_behavior:
  default_on_failure: hold
  default_on_scope_violation: quarantine
  default_on_secret_exposure: freeze_and_escalate
  default_on_network_violation: quarantine
  default_on_missing_witness: hold
  default_on_agent_self_approval: revoke_and_quarantine
  rollback_strategy: none
red_lines:
  offensive_action_requested: false
  live_external_exploitation: false
  hack_back: false
  credential_theft: false
  malware_behavior: false
  covert_persistence: false
  evasion: false
  destructive_action: false
  autonomous_retaliation: false
  third_party_scanning: false
review:
  independent_reviewer_required: false
  executor_reviewer_separation_required: true
  quorum_required: false
  quorum_profile: none
  disagreement_policy: c_review
integrity:
  input_hash: null
  contract_hash: null
  expected_output_hash: null
  canonicalization: yaml_canonicalization
  previous_contract_ref: null
  supersedes_contract_ref: null
```

```
notes:
  human_notes: null
  c_notes: null
  unresolved_issues: []
  assumptions:
    - Authorized project documents only.
```

11.2 Executor patch in sandbox

```
schema_version: cli-agent-task-contract-0.1
contract_id: catc-20260516-130000-schema-patch
task_id: task-schema-patch-001
title: Patch Markdown table formatting in CLI protocol docs
created_at: "2026-05-16T13:00:00Z"
expires_at: "2026-05-16T15:00:00Z"
status: active
requested_by: human_anchor
governing_entity:
  entity_id: ester
  entity_name: Ester
  continuity_ref: null
  authority_scope: repository
human_anchor:
  anchor_id: ivan
  display_name: Ivan
  approval_required: true
  legal_review_required: false
agent:
  agent_id: codex-executor-01
  agent_name: Codex Executor
  provider: openai
  role: executor
  runtime: cloud_cli
  version: null
  capability_profile_ref: null
  trust_level: trusted_limited
objective: Fix Markdown table formatting only in authorized CLI agent protocol documents and
  return a diff plus rollback plan.
risk_class: R2
assertion_class: C-A10
scope:
  repository: cli-agent-protocols
  branch_or_worktree: worktree/cli-schema-formatting
  allowed_paths:
    - docs/cli-agent/
```

```
denied_paths:
  - secrets/
  - memory_core/
  - legal/
  - .env
allowed_commands:
  - markdownlint docs/cli-agent
  - git diff -- docs/cli-agent
denied_commands:
  - git push
  - rm -rf
  - curl
  - wget
allowed_file_globs:
  - "*.md"
denied_file_globs:
  - "*.env"
  - "*.key"
  - "*.pem"
max_files_touched: 5
max_diff_lines: 300
external_targets_allowed: false
external_targets: []
data_policy:
  classification: internal
  secrets_allowed: false
  private_memory_allowed: false
  sealed_material_allowed: false
  legal_privileged_material_allowed: false
  child_data_allowed: false
  raw_witness_evidence_allowed: false
  cloud_upload_allowed: false
  prompt_minimization_required: true
  redaction_required: true
  retention: audit
  output_sanitization_required: true
network_policy:
  mode: none
  allowed_endpoints: []
  denied_endpoints: []
  package_install_allowed: false
  external_fetch_allowed: false
execution:
  sandbox_required: true
  sandbox_id: sandbox-cli-schema-formatting-001
```

```
branch_required: true
container_required: false
dry_run_first: true
max_runtime_minutes: 45
max_retries: 2
max_cost_eur: 3
max_tokens: 200000
stop_on_scope_violation: true
preserve_evidence_before_repair: false
rollback_required_if_partial_write: true
output_required:
  - summary
  - changed_files
  - diff
  - tests_run
  - test_results
  - risk_report
  - rollback_plan
  - commands_run
approval:
  self_approval_allowed: false
  reviewer_required: true
  reviewer_agent_id: gemini-reader-01
  c_gate_required: true
  human_gate_required: false
  legal_review_required: false
  merge_allowed_without_human: false
  publish_allowed_without_human: false
  deploy_allowed_without_human: false
witness:
  witness_required: true
  witness_profile: operational
  witness_event_families:
    - cli_agent.execution
    - cli_agent.review
  append_only_required: true
  hash_required: true
  signature_required: false
memory_gate:
  direct_memory_write_allowed: false
  memory_proposal_allowed: true
  proposed_memory_class: MG-1
  c_review_required: true
  uncertainty_required: true
  quarantine_on_uncertainty: true
```

```
failure_behavior:
  default_on_failure: hold
  default_on_scope_violation: quarantine
  default_on_secret_exposure: freeze_and_escalate
  default_on_network_violation: quarantine
  default_on_missing_witness: hold
  default_on_agent_self_approval: revoke_and_quarantine
  rollback_strategy: revert_patch
red_lines:
  offensive_action_requested: false
  live_external_exploitation: false
  hack_back: false
  credential_theft: false
  malware_behavior: false
  covert_persistence: false
  evasion: false
  destructive_action: false
  autonomous_retaliation: false
  third_party_scanning: false
review:
  independent_reviewer_required: true
  executor_reviewer_separation_required: true
  quorum_required: false
  quorum_profile: executor_tester_auditor
  disagreement_policy: hold
integrity:
  input_hash: null
  contract_hash: null
  expected_output_hash: null
  canonicalization: yaml_canonicalization
  previous_contract_ref: null
  supersedes_contract_ref: null
notes:
  human_notes: Only formatting changes. No semantic rewrite.
  c_notes: null
  unresolved_issues: []
  assumptions:
    - Repository is owned or authorized.
```

11.3 Invalid offensive contract

This example is invalid and must be rejected.

```
schema_version: cli-agent-task-contract-0.1
contract_id: catc-invalid-hackback
task_id: task-invalid-001
```

```
title: Trace and counterattack hostile source
created_at: "2026-05-16T13:30:00Z"
status: active
requested_by: c
governing_entity:
  entity_id: ester
  entity_name: Ester
  continuity_ref: null
  authority_scope: incident
agent:
  agent_id: codex-executor-01
  agent_name: Codex Executor
  provider: openai
  role: executor
  runtime: cloud_cli
  trust_level: trusted_limited
objective: Counterattack the suspected hostile source.
risk_class: RX
assertion_class: C-A10
scope:
  allowed_paths:
    - ./
  denied_paths:
    - secrets/
  allowed_commands: []
  denied_commands: []
  external_targets_allowed: false
data_policy:
  classification: incident_sensitive
  secrets_allowed: false
  private_memory_allowed: false
  sealed_material_allowed: false
  legal_privileged_material_allowed: false
  child_data_allowed: false
  raw_witness_evidence_allowed: false
  cloud_upload_allowed: false
  prompt_minimization_required: true
  redaction_required: true
  retention: audit
  output_sanitization_required: true
network_policy:
  mode: allowlist
  allowed_endpoints: []
  denied_endpoints: []
  package_install_allowed: false
```

```
    external_fetch_allowed: false
execution:
    sandbox_required: true
    branch_required: true
    container_required: true
    dry_run_first: true
    max_runtime_minutes: 1
    max_retries: 0
    stop_on_scope_violation: true
    preserve_evidence_before_repair: true
    rollback_required_if_partial_write: true
output_required:
    - risk_report
approval:
    self_approval_allowed: false
    reviewer_required: true
    c_gate_required: true
    human_gate_required: true
    legal_review_required: true
    merge_allowed_without_human: false
    publish_allowed_without_human: false
    deploy_allowed_without_human: false
witness:
    witness_required: true
    witness_profile: incident
    witness_event_families:
        - cli_agent.incident
        - cli_agent.anomaly
    append_only_required: true
    hash_required: true
    signature_required: true
memory_gate:
    direct_memory_write_allowed: false
    memory_proposal_allowed: false
    proposed_memory_class: MG-Q
    c_review_required: true
    uncertainty_required: true
    quarantine_on_uncertainty: true
failure_behavior:
    default_on_failure: deny_and_quarantine
    default_on_scope_violation: quarantine
    default_on_secret_exposure: freeze_and_escalate
    default_on_network_violation: quarantine
    default_on_missing_witness: hold
    default_on_agent_self_approval: revoke_and_quarantine
```

```
    rollback_strategy: none
  red_lines:
    offensive_action_requested: true
    live_external_exploitation: false
    hack_back: true
    credential_theft: false
    malware_behavior: false
    covert_persistence: false
    evasion: false
    destructive_action: false
    autonomous_retaliation: true
    third_party_scanning: false
  review:
    independent_reviewer_required: true
    executor_reviewer_separation_required: true
    quorum_required: true
    quorum_profile: custom
    disagreement_policy: human_review
  integrity:
    canonicalization: yaml_canonicalization
```

Required result:

```
deny_and_quarantine
```

12. Contract validation workflow

A governance system SHOULD validate contracts in this order:

```
parse
-> structural schema validation
-> red-line validation
-> role/action validation
-> risk/control validation
-> data/cloud validation
-> network validation
-> approval validation
-> witness validation
-> memory gate validation
-> final allow / hold / deny / quarantine
```

13. Default failure mapping

Failure	Required default
schema parse failure	deny_and_quarantine
missing required field	hold
red-line true	deny_and_quarantine
scope missing	hold
denied path touched	quarantine
secret exposure	freeze_and_escalate
unapproved network access	quarantine
self-approval attempt	revoke_and_quarantine
missing witness for privileged task	hold
output contains prohibited data	quarantine
stale contract	hold
expired contract	hold

14. Implementation notes

14.1 Canonicalization

For high-assurance use, the contract SHOULD be canonicalized before hashing.

Recommended sequence:

```
normalize
sort object keys
remove non-semantic whitespace
encode UTF-8
hash
store contract_hash
```

14.2 Contract immutability

An active contract SHOULD NOT be edited in place after execution begins.

Changes SHOULD create a new contract referencing the previous one through:

```
previous_contract_ref
supersedes_contract_ref
```

14.3 Append-only review

Review decisions SHOULD be append-only.

Corrections and reversals SHOULD reference earlier decisions rather than silently replacing them.

14.4 Human-readable pair

Every machine-readable contract SHOULD have a short human-readable task card:

Task:
Agent:
Scope:
Denied:
Network:
Risk:
Review:
Failure default:

15. Conformance levels

Level	Meaning
CATC-0	no task contract discipline
CATC-1	human-readable task contracts only
CATC-2	machine-readable contracts for write tasks
CATC-3	schema validation + semantic validation
CATC-4	witness-bound contracts for privileged tasks
CATC-5	high-assurance canonicalized, hashed, reviewed, and replayable task contract workflow
CATC-X	revoked / invalid / red-line failure

16. Open issues

ID	Issue	Required action
0I-001	Stable \$id URL	Replace placeholder schema URL with final repo path.
0I-002	Canonicalization profile	Decide JSON vs YAML canonicalization.
0I-003	Signature profile	Define signing format for high-assurance contracts.
0I-004	Agent registry reference	Create CLI_Agent_Registry_Profile_v0_1.md.
0I-005	Witness event binding	Create CLI_Agent_Witness_Event_Profile_v0_1.md.
0I-006	Memory gate binding	Create CLI_Agent_Memory_Gate_Profile_v0_1.md.
0I-007	Runtime validator	Create reference validator for schema + semantic rules.
0I-008	Provider profiles	Define local, OpenAI, Google, and other provider-specific capability envelopes.
0I-009	High-risk task examples	Add more R4/R5 examples after root protocol review.
0I-010	Repo placement	Decide final GitHub path and package index integration.

17. Closing rule

A CLI agent task contract is not bureaucracy.

It is the boundary between useful executable help and uncontrolled operational drift.

Final rule:

If the task cannot be scoped, it cannot be delegated.

If it cannot be reviewed, it cannot be integrated.

If it crosses a privileged boundary, it must be witnessed.